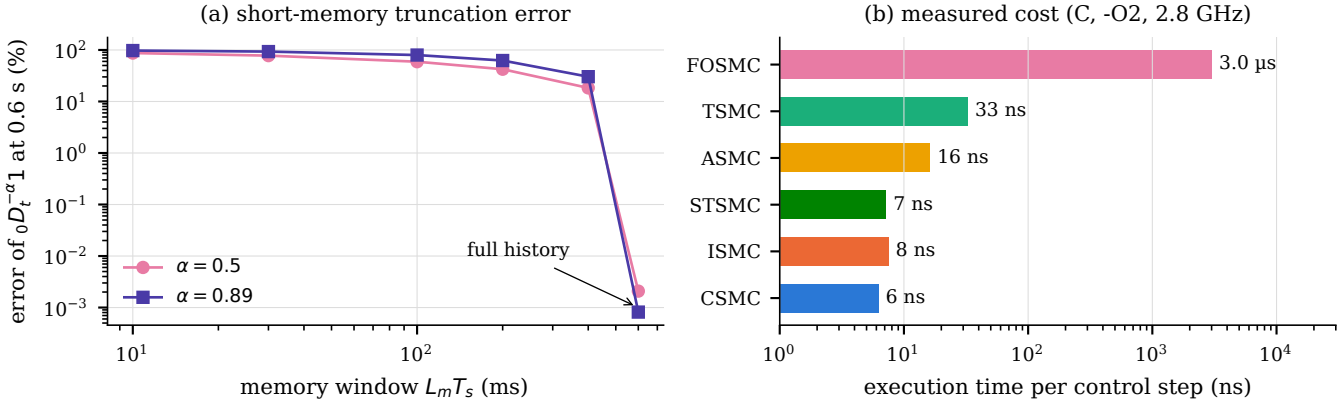
(a) short-memory truncation error
error of ${}_0D_t^{-\alpha}1$ at 0.6 s (%)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
$\alpha = 0.5$
$\alpha = 0.89$
full history
10^{1}
10^{2}
memory window $L_m T_s$ (ms)
(b) measured cost (C, -O2, 2.8 GHz)
FOSMC
TSMC
ASMC
STSMC
ISMC
CSMC
3.0 µs
33 ns
16 ns
7 ns
8 ns
6 ns
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
execution time per control step (ns)